



Investigations on performance of single-phase immersion cooling system

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ABSTRACT

Cooling of data centers is crucial from the point of view of energy consumption and zero downtime of data centers. The present study examines the heat transfer behavior for a single-phase immersion cooling system for dielectric fluid (FC40) and dielectric oil (PAO-6) in a 1 U server. Thermal resistance and the temperature of the server are considered to study the system's thermal performance at various system parameters. Parameters varied are inlet/outlet configuration, bypass of the fluid (a gap between the heat sink and the fluid tank), imposed suction fans, fin pitch of the heat sink, fluid flowrate, fluid inlet temperature, and heat load. T-type configuration of inlet/outlet outperforms Z-type configuration. The performance of the system is reduced monotonically with the increase in the gap spacing between the heat sink and tank due to the increased bypass of fluid. The suction created downstream of the heat sink by deploying two fans in series improves the performance appreciably. The effect of the fin pitch of the heat sink on the performance of the system is different for both fluids. For PAO-6, the fin pitch of 2.2 mm yields better performance than the dense pitch of 1.2 mm whereas FC-40 shows the opposite trend. Thermal resistance considerably decreases for PAO-6 with an increase in fluid flowrate from 1 to 3 LPM. Thermal resistance decreases whereas the temperature of the server increases when the fluid inlet temperature increases from 15 to 35 °C. A similar effect is reported when the heat load is increased from 200 to 600 W. Numerical investigations are also carried out for studying to explain the flow field and temperature distribution in case of bypass of fluid.

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1. Introduction

The recent developments in technologies such as the fifth generation (5 G) mobile communication technology, artificial intelligence technology, the internet of things, big data technology, and automation have boosted the demands of IDCs (internet data centers) which contain physical or virtual servers to facilitate high-performance computing, complex transactions, huge storage, and many more processes. The information technology sector is expected to consume approximately 20% of the world's energy by 2030 [1]. Energy consumption in DCs is one of the main concerns as it has increased globally to 205 TWh, accounting for about 1% of the world's total electricity consumption in 2018 [2]. Cooling systems consume about 40% of the total energy bill in the DC [3,4]. In this regard, effective heat dissipation from electronic chips in various servers in the DC has been a challenge amid the growing demand for high power density and higher clock frequency of microprocessor chips. Modern DCs typically require a large dissipation

of heat (5–15 MW). To reduce the CO₂ footprint, cooling efficiently plays a crucial role from the perspective of energy consumption. The advances in the thermal management of DCs are reviewed by several researchers Fulpagare and Bhargava [5], Alkharabsheh [6], Khalaj and Halgamuge [7], Kheirabadi and Groulx [8], Daraghme and Wang [9], and Chu and Wang [10]. Multi-scale thermal management with different length scales (viz. chip, server, chassis, rack, room, and plenum level) involved in the DC is briefed by Joshi and Kumar [11]. Analysis of a bare heat-generating chip subject to pure conduction and convection cooling boundary conditions is presented by Gurram et al. [12].

Micro fins are widely used for various applications such as cryogenic coolers with two-phase liquid nitrogen flows [13], boiling heat transfer [14], and single phase fluid flows [15]. Bar-Cohen et al. [16] reviewed recent investigations on embedded cooling for wide bandgap power amplifiers. Experimental investigation on natural circulation in connected vertical rectangular loops is presented by Misale et al. [17]. Numerical investigations on thermal performance of a mini natural circulation loop using lattice Boltzmann method is reported by Bocanegra et al. [18]. Air cooling has been the traditional and mature method that requires large air conditioning systems, porous floors, and air distribution units

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Nomenclature

Q_I	input power (W)
R	thermal resistance ($^{\circ}\text{C W}^{-1}$)
T	temperature ($^{\circ}\text{C}$)
T_{in}	inlet temperature of the fluid ($^{\circ}\text{C}$)
T_{out}	outlet temperature of the fluid ($^{\circ}\text{C}$)
V	voltage (V)
I	current (A)

Acronyms

ASHRAE	American society of heating, refrigerating, and air-conditioning engineers
CFD	computational fluid dynamics
CPU	central processing unit
DC	data center
EXP	experimental
GPM	gallons per minute
LPM	liters per minute
PAO	polyalphaolefin
PCB	printed circuit board
PUE	power usage effectiveness
RAM	random-access memory
RNG	re-normalization group
TIM	thermal interface material
TTV	thermal test vehicle

for efficient cooling of each cabinet in the DC. Limitations of air cooling such as smaller cooling capacity ($< 37 \text{ W cm}^{-2}$) are addressed [19], especially for high-performance computing. Smaller thermal conductivity and specific heat of air lead to an unacceptable gigantic surface area, unaffordable fan power, and unbearable noise across the heat sinks. This is because the acoustic noise is empirically reported to be proportional to the fifth power of fan speed [11]. Liquid cooling is a more energy-efficient alternative to conventional air cooling as the former offers a cooling capacity as large as 202 W cm^{-2} [20]. A detailed comparison between a closed and open-loop liquid cooling system is provided by Levin et al. [21]. Liquid cooling can be made available either via indirect (e.g. cold plate) or direct contact (e.g. immersion cooling). In this study, efforts are focused on single-phase immersion cooling. The single-phase immersion cooling technique essentially employs non-conductive fluid with sufficiently large boiling points to ensure that there is no phase change. The electronic equipment is directly submerged in the dielectric fluid for heat dissipation.

The immersion cooling technique for cooling DCs is reviewed by Kuncoro et al. [22]. A two-phase immersion cooling system, compared to a single-phase system, offers a larger cooling capacity owing to the latent heat of vaporization. However, the former demands the fluid with a boiling point that is often smaller than the maximum allowable temperature of electronic equipment. Baker [23] provided early experimental investigations in 1973 on liquid immersion cooling focusing on the investigation of heat source size and substrate conductivity. Convection heat transfer is reported to reduce the size of the heat source and substrate conductivity had a negligible effect on the overall heat transfer performance for a larger heat source. Thermodynamic, economic, and thermo-economic assessments of both single-phase and two-phase immersion cooling systems with cooling capacities in the range of 3.2–27.6 kW and 6.8–15.9 kW, respectively were reported [24]. Replacement and energy cost are two key issues that must be addressed carefully for the two-phase and single-phase immersion cooling systems respectively. Both single-phase and two-phase immersion cooling systems are reported to achieve an appreciable energy-saving when compared to the air-cooled system.

Upon selection of the liquid cooling, one of the major concerns is the potential for leakage. Water or water-based liquids such as ethylene glycol can damage electronic equipment when leakage occurs. To tailor this problem, dielectric fluids or mineral/synthetic oils with single-phase immersion cooling systems can be employed to eliminate the electrical hazard. However, it is critical to select a proper dielectric fluid with a higher flash point to avoid the risk of fire. Safety guidelines are periodically reported by the American society of heating, refrigerating, and air-conditioning engineers (ASHRAE) for working fluids [25]. Ramdas et al. [26] provided useful information through experimental investigations about changes in the material properties of printed circuit boards for immersion in the dielectric oil EC 100. Young's modulus and stiffness were decreased after immersion while reliability is unaffected. Shah et al. [27] reported that immersion cooling systems may offer improved reliability when compared to air cooling systems. This is because the immersion system contains no moving parts, and offers a lower operating temperature with no oxidation or corrosion of electric contacts of the printed circuit boards (PCBs). Moreover, no potential electrostatic discharge, no sensitivity to environmental impurities, and moisture were seen in the immersion system. Luo et al. [28] reported the optimization of thermophysical properties of silicon carbide oil-based nanofluid (specific heat and thermal conductivity) for submerged cooling of DCs with Reynolds numbers ranging from 250 to 3000. Appreciable enhancement of heat transfer coefficient and lower and uniform temperatures are reported for nanofluids compared to base mineral oil.

Experimental and numerical investigations are widely reported in the literature. Effects of the flowrate and inlet temperature of the mineral oil on the thermal performance of the single-phase immersion cooling system for a single server are studied experimentally by Eiland et al. [29]. The possibility to utilize inlet temperatures of oil up to 45°C and power usage effectiveness (PUE) ranging from 1.03 – 1.17 for the cooling system was concluded. Bansode et al. [30] experimentally investigated the thermal performance of the system with the server being immersed in a single-phase dielectric fluid over a wide range of temperatures (25 to 55°C) inside an environmental chamber for a prolonged period (144 h). The average core temperature of the server increased drastically with the environmental temperature. However, the power consumption of the server remains unchanged. Bansode et al. [31] studied further the same under extreme environmental conditions experimentally from -20 to 10°C at 100% relative humidity and from 20 to 55°C at 50% relative humidity. No failure of the system is reported and transient variation in the average consumed power of the server was insignificantly low (about 20 W). Gandhi et al. [32] examined the thermal performance of single-phase oil immersion cooling for a high-performance computing server against air cooling using computational fluid dynamics (CFD, 6SigmaET). The inlet oil temperature and flowrate were varied to arrive at the optimized solution subject to synthetic oil (EC – 100) and mineral oil. Shinde et al. [33] experimentally investigated the effects of the oil flowrate and its inlet temperature on the thermal performance of a single-phase immersion cooling system and the power consumption of the server. The reported PUE for the cooling system ranged from 1.03 – 1.17. Compared to air cooling, the reduction in server power is 16.83% for oil cooling. Cheng et al. [34] investigated the effect of the heat sink's material and the oil flowrate on the performance of single-phase immersion cooling with forced convection. It is concluded that both aluminum and copper heat sinks offer comparable performance. Although it is found that temperature distribution in the heat sink is asymmetric, a larger flowrate offers a more uniform temperature distribution. A liquid immersion cooling technology with natural convection is proposed through numerical and experimental investigations [35] with silicone oil, soybean oil, and perfluorocarbon-structured liquid as dielectric fluids.

Table 1

Summary of the investigations reported in the literature on single-phase immersion cooling systems.

Refs.	Dielectric oil	Heat load (W)	System parameters/variables	Remarks
[29]	Mineral Oil	95	An oil flowrate of 0.5–2.5 LPM and an inlet temperature of oil (T_{in}) of 30–50 °C	Possibility to utilize T_{in} up to 45 °C and partial PUE in the range of 1.03 – 1.17
[30]	Opticool 872,552	100–110	T_{in} : 30–60 °C and environmental temperature: 25–55 °C	The average temperature of the server increased drastically with the environmental temperature however, the power consumption of the server remains unchanged.
[32]	EC-100	500	T_{in} : 15–25 °C and oil flowrate of 0.5–3 GPM	The optimized solution is obtained using CFD on 6SigmaET
	Mineral Oil		T_{in} : 25–45 °C and oil flowrate of 0.5–1 GPM	
[33]	Mineral Oil	95	T_{in} : 25–45 °C and oil flowrate of 0.5–1 GPM	Possibility to utilize T_{in} up to 45 °C and partial PUE in the range of 1.093 – 1.1
[36]	Mineral Oil and EC-100	115	T_{in} : 30–50 °C and oil flowrate of 0.5–2.5 LPM	Plate-fin, pin fin, and parallel plate heat sinks offer similar performance.
[37]	Mineral Oil and EC-100	115	T_{in} : 30–50 °C and oil flowrate of 0.5–2.5 LPM	System performance is reported the best for a 1.5 U server at 40 °C and a flowrate of 1 LPM with EC-100
[31]	Opticool 872,553	100–110	T_{in} : 30–60 °C and environmental temperature: –20 to 55 °C	No failure of the system is reported and transient variation in the average consumed power of the server was insignificantly low (about 20 W)
[34]	Novec 7100	95	Oil velocity: 0.4 – 1.2 m s ^{–1} and material of heat sink	Temperature distribution in the heat sink is asymmetric, however, the larger flowrates offer a uniform one, and no effect of the heat sink's material on performance
[39]	Mineral Oil + Al ₂ O ₃	130	The concentration of Al ₂ O ₃ , design of heat sink, T_{in} : 25–45 °C, and oil flowrate of 1–3 LPM	Pin fin heatsink, oil flowrate of 1 LPM with 5% concentration of Al ₂ O ₃ with T_{in} : 25 °C is the most efficient solution
[35]	Silicone oil, soybean oil, and perfluorocarbon structured liquid	7–14 kW	Water - T_{in} : 15–35 °C and flow velocity: 0.005–1 m s ^{–1}	A PUE of 1.04 is demonstrated and a perfluorocarbon-structured oil is reported as the most suitable oil. A numerical investigation with a heat load of 440 W is conducted.

Yet, a PUE of 1.04 is demonstrated and a perfluorocarbon structured fluid (Fluorinert) is reported as the most suitable fluid. The performance of various heat sink designs is compared by numerical investigations for single-phase liquid immersion cooling systems [36]. No significant performance improvement for the plate-fin heat sink and pin-fin heat sink is reported when compared to the conventional parallel plate heat sink. The comparison between the performances of oil-cooled servers with different open rack units is investigated numerically and the optimum solution is provided [37]. Thermo-hydraulic performance of a natural circulation mini-loop using distilled water and nanofluid (distilled water with nanoparticles of Al₂O₃) is compared experimentally by Misale and Garibaldi [38], and no appreciable difference in performance was reported. The effects of the concentration of Al₂O₃, heat sink, fluid inlet temperature, and fluid flowrate were studied numerically and the optimum solution is provided by Niazmand et al. [39]. Levin et al. [21] reported that the maximum temperature of the field-programmable gate array can be reduced to 55 °C upon using the immersion cooling system. A summary of the investigations reported in the literature on single-phase immersion cooling systems is summarized in Table 1.

From the table, heat loads studied in the literature range from 90 to 130 W except [32,35]. Thus, there is a need to investigate the performance experimentally with larger heat loads. The effect of various dielectric fluids and system parameters such as fluid flowrate and inlet temperature of the fluid is mainly reported in the literature. However, due to the comparatively high viscosity along with its strong dependence on temperature, flow maldistribution with considerable bypass may occur. Yet, the consid-

erable variation of viscosity subject to temperature variation may also change the thermal performance. These important parameters affecting the overall performance of single-phase immersion cooling are not well elaborated yet. The present work is aimed at filling the above-mentioned research gap. PAO-6 is used in the present work as dielectric fluid and the performance is compared with FC-40 as dielectric fluid investigated in the previous work [40].

2. Experimental methodology

The schematic diagram of the experimental rig of the single-phase liquid immersion cooling system is shown in Fig. 1. The data server (1 U) is immersed in the oil tank. An oil pump is used to supply a constant oil temperature from the thermostatic bath. The data server consists of an Intel-based thermal test vehicle (TTV) and a 3D-printed random-access memory (RAM)-like structure mounted on the PCB. Heat load from the RAM is not considered in the present work as it is negligible compared to the central processing unit (CPU). TTV that is powered by an external DC power supply can be referred to as a simulator of a CPU capable of producing the equivalent heat load.

The corresponding plate fin heat sink mounted on the CPU and RAM attached to the PCB is shown in Fig. 2. The lower and upper pins of the PCB are used to connect the DC power supply for heating the TTV. Thermal resistance at the interface between the copper heat sink and TTV is reduced by applying thermal interface material (a thermal paste: TC-5888 with thermal conductivity of 5.2 W m^{–1} K^{–1}). The actual experimental rig is shown in Fig. 3 which depicts the 1 U server immersed in the oil tank. The entire

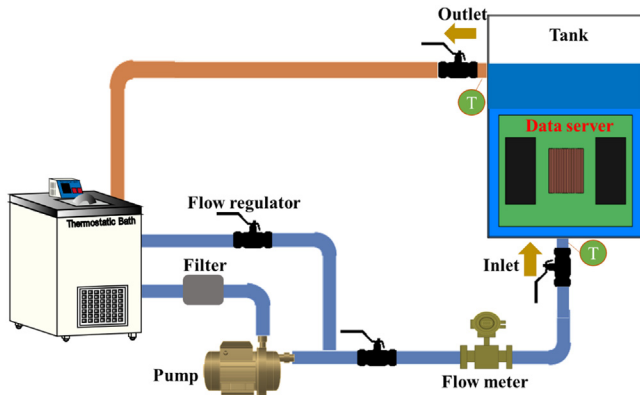


Fig. 1. Schematic diagram of the experimental rig.

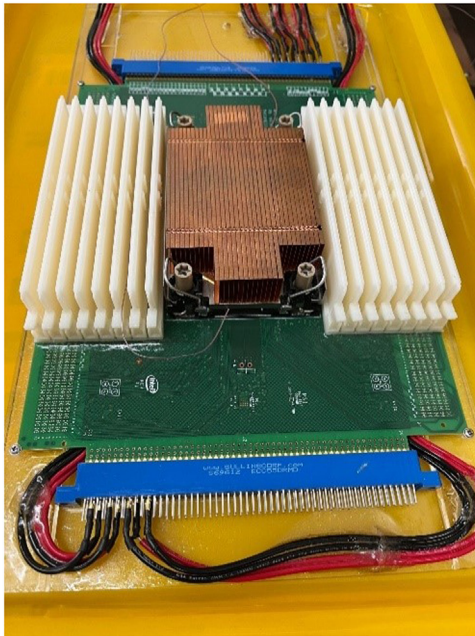


Fig. 2. Equivalent data server fixed on PCB.

experimental setup is well insulated to minimize heat loss. A thermostatic bath (Make: Firstek B20 with 1.9 kW cooling power) is used to control the inlet temperature of fluid into the 1 U server. The flowrate of the oil is controlled by the flow regulator (bypass) and is measured through an oil flow meter (Make: BNC Industrial-BH20-0F5-BSP with the range of 0.33 to 21.67 LPM). Experiments are conducted till a steady state is reached for all the desired parameters to be recorded. A data acquisition system (MX 100) is used to ensure a steady state.

Two T-type thermocouples are fixed on TTV surfaces to measure the temperature of the server (T_{TTV}). The temperatures of the oil are measured at the inlet (T_{in}) and the outlet (T_{out}) of the tank using T-type thermocouples. Voltage (V) and current (I) from the DC power supply (6230K-150) are recorded to evaluate the input power ($Q_I = VI$). The overall thermal resistance of the system is given below.

$$R = \frac{T_{TTV} - T_{in}}{Q_I} \quad (1)$$

The effects of various parameters on the performance of the system are investigated, including the effects of inlet/outlet, bypass, fan arrangement (induction or forced type), a heat sink (fin pitch), oil flowrate, oil inlet temperature, and heating load. The



Fig. 3. Image of immersion cooling system.

Reynolds number based on fin spacing within the heat sink for FC-40 falls around 10–70 (depending on the flowrate, temperature, and fin pitch). While for PAO-6, the Reynolds number is around only 0.2 to 1. At first sight, it looks like laminar flow prevails. However, the experimental system contains heat sink, RAM, fan, contraction inlet and expansion outlet, and fans. Yet local variation in viscosity subject to change of heat load and temperature are also quite pronounced. Hence, the flow field is kind of complex. The system parameters studied in the present work are described in Table 2.

With the accuracy of the T-type thermocouple, voltmeter, and ammeter as 0.1 °C, $\pm 0.05\%$, and $\pm 0.1\%$ respectively, the maximum uncertainty in measuring the temperature of the server and evaluating the thermal resistance of the system is 0.71 and 0.72% respectively. Precision error with suggested degree of confidence for experimental measurements [41] and bias error are considered for evaluating the uncertainty.

PAO-6 is a synthetic oil (poly-alpha-olefin), which is often used as the main base oil for engines, heavy engines, gearboxes, and other industrial lubricants. Its flash point is considerably high (246 °C) and its safety behavior is superior to other fluids. The performance of PAO-6 is then compared with dielectric fluid FC-40. For both dielectric liquids, variations in viscosity, density, thermal conductivity, and specific heat with temperature are shown in Fig. 4. The viscosity of both fluids reduces with the rise of temperature significantly. Reduction in the viscosity for FC-40 and PAO-6 over the range of fluid inlet temperature (15–35 °C) is 38% and 57%, respectively. Reduction in viscosity with an increase in temperature can be attributed to an increase in molecular kinetic energy and decreased attraction forces between the molecules. The viscosity of PAO-6 is markedly larger than that of FC-40 over the working range in this studied range (> 15 °C). Pumping power (hence operation cost) is largely dependent on pressure drop which is also strongly related to the viscosity. Note that the temperature across the heat sink may be quite different at the base and tip of the heat sink, thereby yielding velocity variation alongside the heat sink. A monotonic decrease in the pumping power, which is qualitatively similar to the decrease in viscosity with fluid temperature, is reported in [29]. There is a considerable difference in properties

Table 2
System parameters studied in the present work.

Parameter	Values/description	Illustration	Remarks
Inlet/outlet configuration	T, Z, and U type		The inlet/outlet configuration can vary in practice according to the available actual server. The flow pattern inside the tank may strongly depend on the positions of the inlet(s) and outlet(s).
Bypass of the fluid (a gap between the heat sink and the fluid tank)	0, 2, and 5 mm		The bypass may significantly offset the thermal performance upon immersion cooling, this phenomenon may be accentuated at lower flowrates with multiple heat sinks design. An acrylic strip is used to avail different gaps
Suction fans	2 fans in the series		The high viscosity of fluid limits the flowrate to pass the heat sink. Two fans (13,000 rpm) with a series connection are deployed at the exit of the heat sink.
Fin pitch of the heat sink	Heat sinks A and B with fin pitch of 1.2 and 2.2 mm		Fin pitch affects the heat transfer coefficient on the fluid side and also the pressure drops across the heat sink
Fluid flowrate	1, 2, and 3 LPM	–	One of the primary parameters affecting the heat transfer With a considerable change in the viscosity of the fluid with the temperature, the associated heat transfer performance may also vary appreciably. The temperature of the server directly varies with the heat load
Fluid inlet temperature	15, 25, and 35 °C	–	
Heat load	200–600 W in the step of 100	–	

such as density, thermal conductivity, and specific heat of PAO-6 as compared to FC-40. The density of PAO-6 is 2–2.3 times smaller than that of FC-40 at the same temperature, whereas, thermal conductivity and specific heat are 2.2 times larger. Variation in density, thermal conductivity, and specific heat over the working range of the temperature is only minor (< 6%) when compared to the viscosity.

3. Results and discussion

The effect of various parameters such as the inlet/outlet configuration, bypass of the fluid, assisted fan, fin pitch of the heat sink, fluid flowrate, fluid inlet temperature, and heat load, on the performance of the single-phase liquid immersion cooling system is presented in this section. The effect of the fin pitch of the heat sink is included while reporting the effect of other system parameters.

3.1. Inlet/outlet configuration

A comparison of the system performance in terms of thermal resistance and the maximum case temperature of the server subject to various inlet/outlet configurations (T, Z, and U) is presented

in Fig. 5. The thermal resistance for U and Z-type configurations is comparable, meaning that the outlet location of tank imposes negligible influence of the overall performance. Note that the T-type configuration offers slightly better performance than those of Z or U configurations with a smaller thermal resistance (12.6% and 5.1% less for FC-40 and PAO-6 respectively) and a lower case temperature (4.9% and 2.3% less for FC-40 and PAO-6 respectively). This improved thermal performance of the system for T-type configuration is due to the inlet fluid flowrate being in front of the heat sink, thereby entraining more fluid across the heat sink.

3.2. Bypass of the fluid

Effect of the bypass on the thermal resistance and temperature of server for both heat sinks is shown in Fig. 6.

For both heat sinks, the presence of a gap reduces the cooling performance of the system. Both the overall thermal resistance and the temperature of the server increase as the gap increases from 0 to 5 mm due to more bypass of the fluid. Although bypass of fluid is important to minimize thermal shadowing in case of multiple heat sinks, unnecessary bypass of oil should be avoided for efficient heat dissipation. A monotonic increase in both thermal resistance and temperature of server with the gap is qualitatively

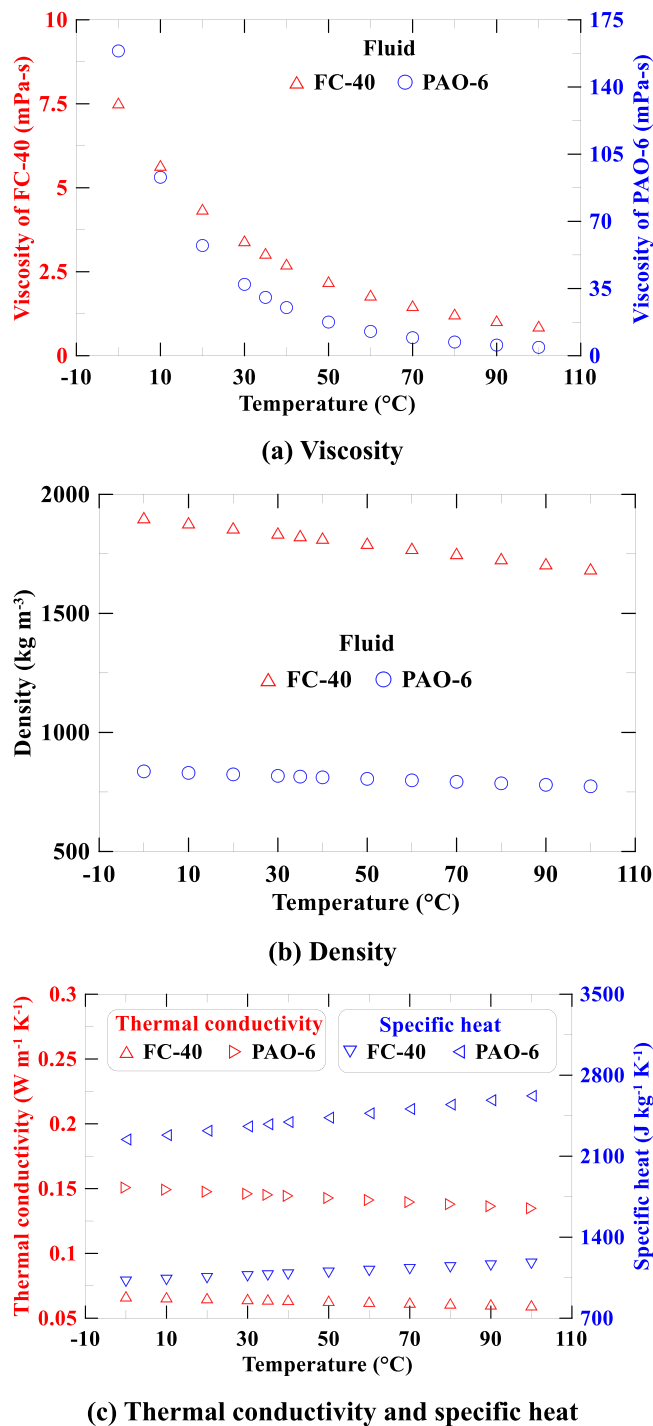


Fig. 4. Variation in viscosity, density, thermal conductivity, and specific heat of FC-40 and PAO-6 with temperature.

similar and more significant for PAO-6 compared to FC-40. Thermal resistance is increased by 13.6% and 12.3% for heat sinks with fin pitch of 1.2 and 2.2 mm, respectively for PAO-6 oil whereas the same is observed as 6.9% and 2.9% for FC-40. Note that the PAO-6 offers much higher viscosity, and accordingly a much higher resistance for fluid flowrate flowing across the heat sink, thereby resulting in a more severe bypass. Variation in thermal resistance or the temperature of the server with an increase in the bypass for both fluids is comparable for both heat sinks. However, it is interesting to note that the effect of fin pitch of the heat sink on the thermal performance is different for both fluids. Heat sink A offers

Table 3
Boundary conditions imposed.

Inlet velocity	0.28 m s ⁻¹ (corresponding to 3 LPM)
Inlet temperature	308.15 K (35 °C)
Heat flux at the surface of the server	74,148.22 W m ⁻² (corresponding to 300 W)
Outlet	Pressure outlet
Other walls	Adiabatic

roughly 36–38% more thermal resistance compared to heat sink B with PAO-6 as a working fluid whereas, with FC-40, heat sink B offers around 9–14% more thermal resistance compared to heat sink A. This can be attributed to the large difference in the viscosity of both fluids. Heat sink A with a fin pitch of 1.2 mm may offer PAO-6 with a larger flow resistance for fluid to pass the fin array and yield an inferior performance when compared to Heat sink B with a fin pitch of 2.2 mm. Note that the thermal conductivity is roughly 2700 times smaller compared to copper.

3.2.1. Numerical investigations

Numerical investigations are conducted to study the bypass effect to avail the flow field and temperature distributions. Investigations are carried out using the commercially available CFD package Ansys Fluent (version: 2020R2). The structured mesh with a maximum aspect ratio of 32.79 is opted for in the present work. The average and maximum skewness observed are 0.26 and 0.61 respectively. K-epsilon re-normalization group (RNG) model is selected as a turbulence model with standard wall functions as near-wall treatment. The imposed boundary conditions are listed in Table 3. Grid independence analysis tested for the maximum temperature of the server is shown in Fig. 7. Number of grids equal to 7.18 million is selected for further study as there is a marginal difference in the result after increasing the number of grids beyond this number. Although grid independence test is carried out for the maximum temperature of the server, the same number of grids (7.18 million) are used for studying other properties. Simulation is carried out for studying the effect of bypass of fluid with PAO-6 oil, T-type configuration, no suction fans, heat sink A with a fin pitch of 1.2 mm, the flowrate of 3 LPM, the fluid inlet temperature of 35 °C, and a heat load of 300 W. A comparison between CFD and experimental results for bypass of fluid is shown in Fig. 8 for both fluids. A monotonic increase in the thermal resistance of the system is observed with an increase in the bypass of fluid similar to experimental results. The maximum deviation of the numerical simulation relative to the experimental results is 14.25%. Results of numerical investigations facilitate further possible investigations with significant accuracy without performing experiments.

The effect of the bypass of fluid on the flow field and temperature distribution in the system is studied in the present work. Velocity and temperature contours are plotted on both vertical and horizontal planes through the center of the server as shown in Figs. 9 and 10 respectively presenting the effect of the bypass. Considerable difference in both the magnitude of velocities of fluid and temperatures in the system is observed between no bypass and bypass of fluid conditions. In the absence of bypass of fluid, larger velocities of fluid are observed before the server whereas, in the case of bypass of fluid, fluid bypassing the server with as high as 0.05 m s⁻¹ at the horizontal center plane of the server is reported. Temperature distribution downstream the server is observed due to a relatively colder bypassed fluid stream mixing downstream with the fluid passed over the server (relatively at a larger temperature due to heat dissipation). In case of no bypass of fluid, the almost uniform temperature of the fluid is reported downstream of the server.

Velocity and temperature contours at the horizontal center plane of the server indicate larger velocities of fluid over the server

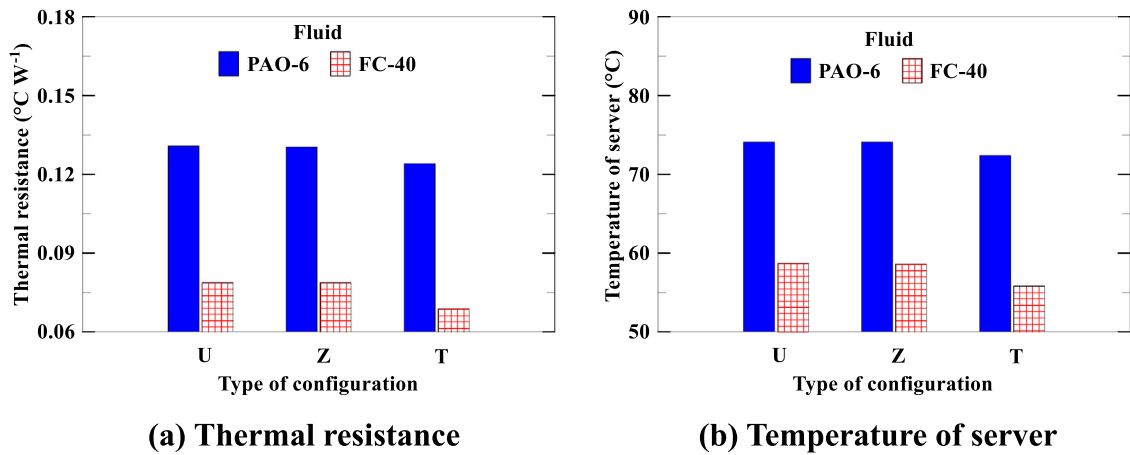


Fig. 5. Comparison of thermal resistance and temperature of the server for T, Z, and U type configurations (No bypass of fluid, no suction fans, heat sink A with fin pitch of 1.2 mm, fluid flowrate: 3 LPM, fluid inlet temperature: 35 °C, and heat load: 300 W).

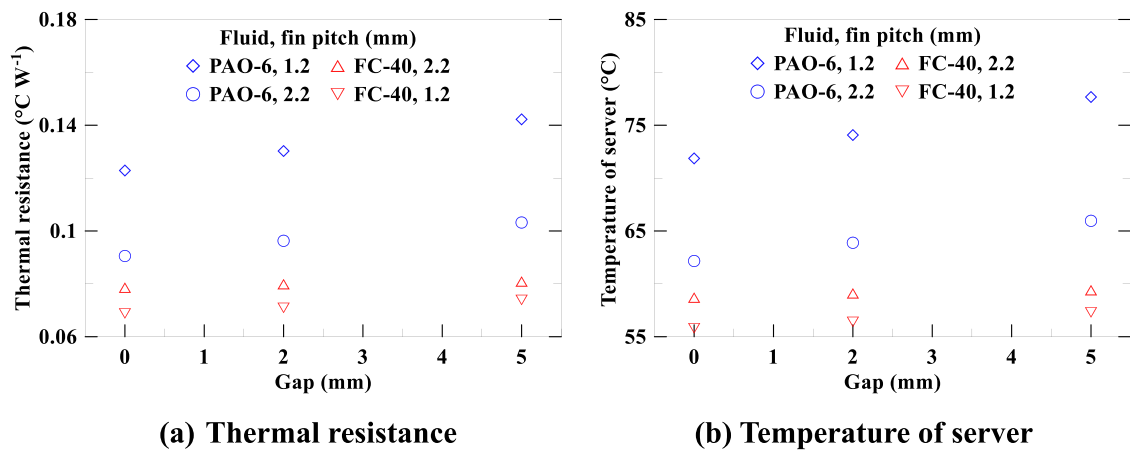


Fig. 6. Effect of bypass on thermal resistance and temperature of server for both heat sinks (T-type configuration, no suction fans, fluid flowrate: 3 LPM, fluid inlet temperature: 35 °C, and heat load: 300 W).

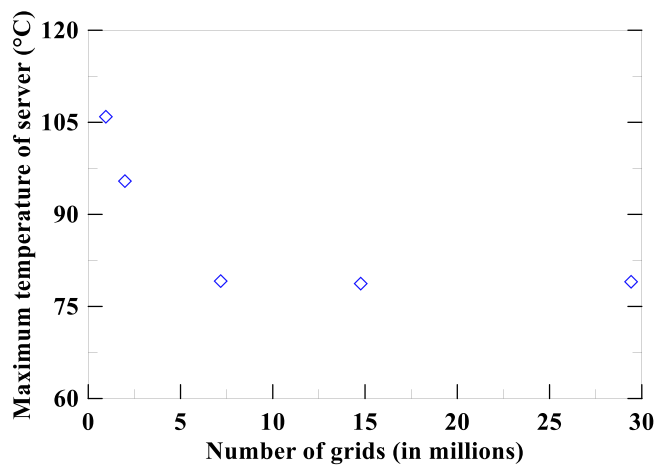


Fig. 7. Mesh independence analysis.

in case of no bypass resulting in improved forced convection and hence smaller temperatures of the server compared to the case of bypass of fluid.

3.3. Effect of suction fans downstream of the heat sink

The effect of adding fans at the exit of the heat sink to entrain more fluid into the heat sink is studied. Both T and Z-type configurations are examined and the results are shown in Fig. 11.

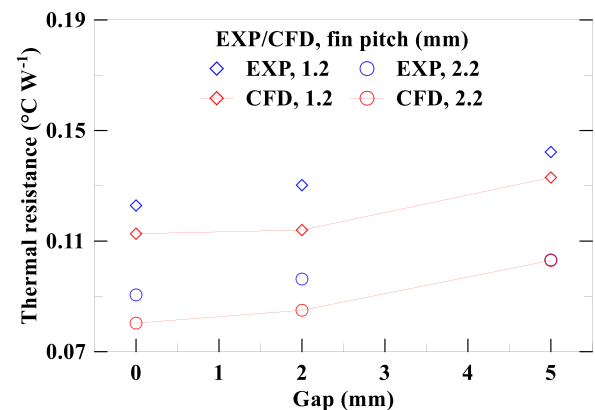


Fig. 8. Comparison between CFD and experimental results for bypass of PAO-6 oil (T-type configuration, no suction fans, heat sink A with fin pitch of 1.2 mm, flowrate: 3 LPM, fluid inlet temperature: 35 °C, and heat load: 300 W).

The fans improve the performance of the system for both dielectric fluids and both configurations. Reduction in both thermal resistance and temperature is more significant for PAO-6 compared to that for FC-40 owing to the improved forced convection of high-viscosity oil PAO-6. For T-type configuration with FC-40 as a working fluid, the thermal resistance of the system and temperature of the server decreases by around 9.3% and 3.8%, respectively when two fans are used compared to no fan, whereas for PAO-6, the re-

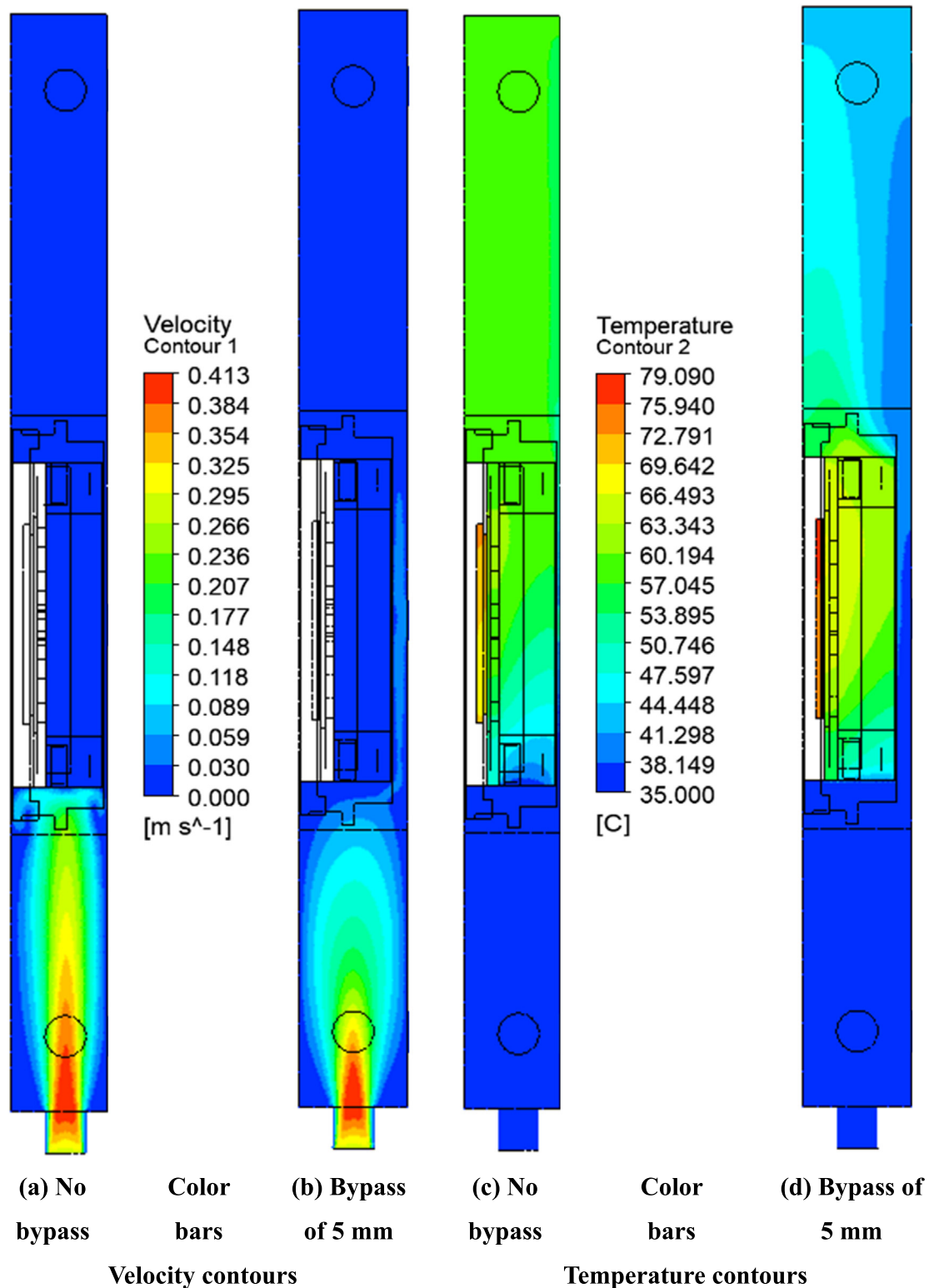


Fig. 9. Effect of bypass on temperature and velocity distribution – contours on the center vertical plane (T-type configuration, no suction fans, heat sink A with fin pitch of 1.2 mm, flowrate: 3 LPM, fluid inlet temperature: 35 °C, and heat load: 300 W).

duction in the same arrangement is 15.6% and 8.3%, respectively. For Z-type configuration with FC-40 as a working fluid, the thermal resistance of the system and temperature of the server is decreased by around 13% and 5.3% respectively when two fans are used, whereas, for PAO-6, the reduction in the same layout is 16.7%

and 8.8%, respectively. Effect of placing suction fans downstream of the heat sink is comparable for both T-type and Z-type configurations, however, the effect of fans is considerably large for PAO-6 compared to FC-40 owing to the larger viscosity of the former that offers greater resistance to the flow. Thus, imposing suction

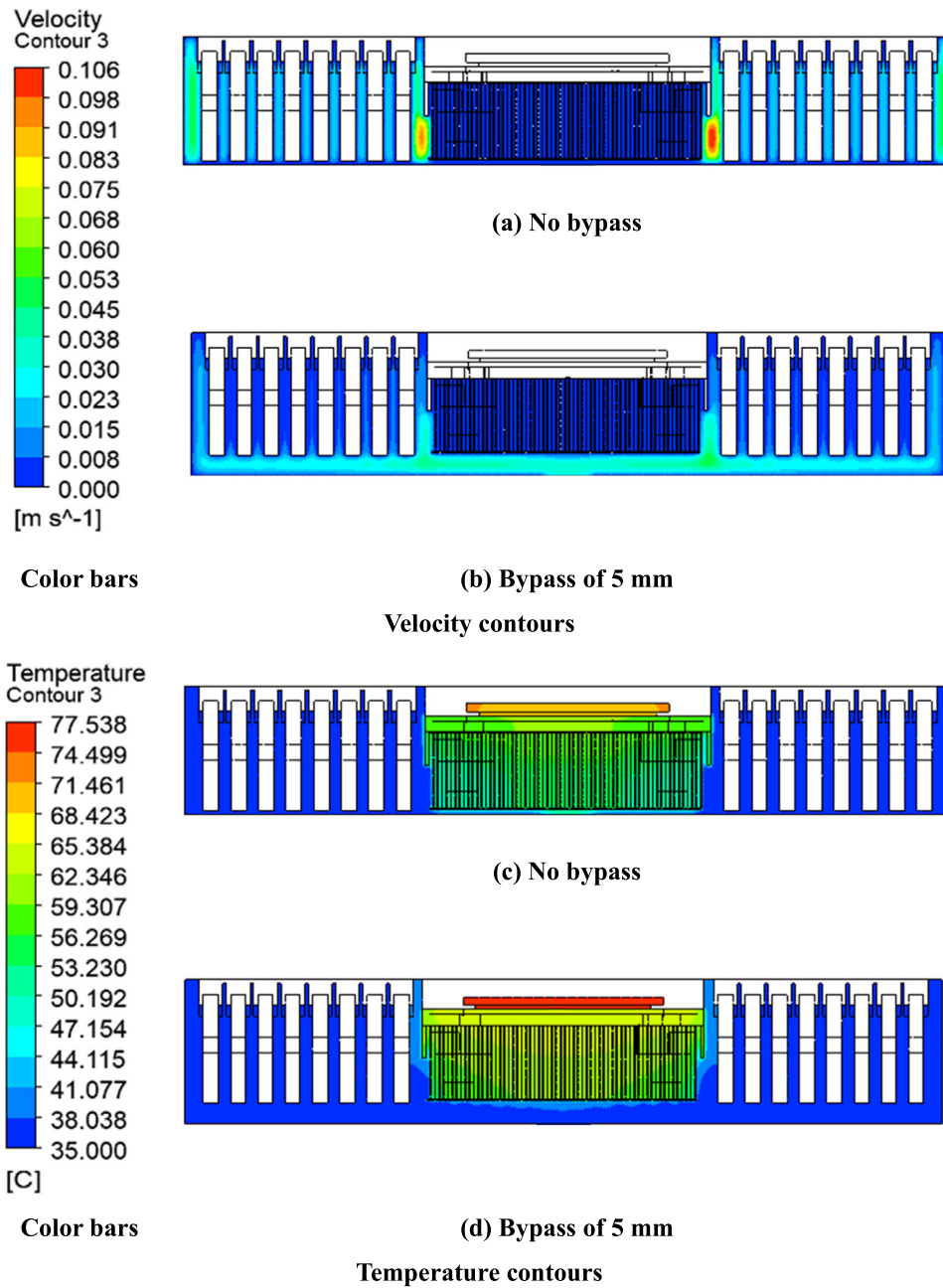


Fig. 10. Effect of bypass on temperature and velocity distribution – contours on the center horizontal plane (T-type configuration, no suction fans, heat sink A with fin pitch of 1.2 mm, flowrate: 3 LPM, fluid inlet temperature: 35 °C, and heat load: 300 W).

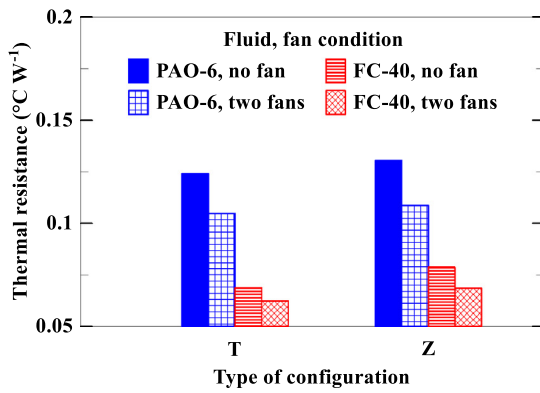
fans can significantly improve the thermal performance of the system that employs the dielectric fluid with large viscosity. Heat sink A provides roughly 34–35% more thermal resistance compared to heat sink B with PAO-6 as a working fluid whereas, with FC-40, heat sink B offers around 6–9% more thermal resistance compared to heat sink A. This can be attributed to a large difference in the viscosity of both fluids.

3.4. Fluid flowrate

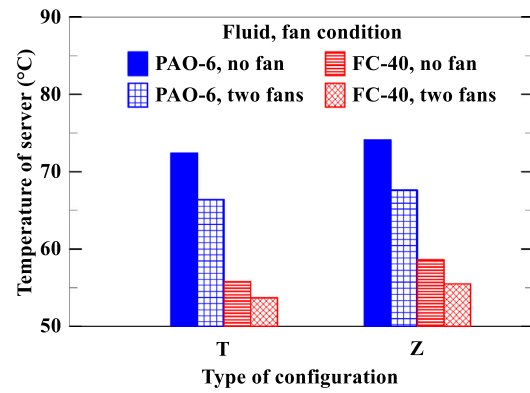
Effect of fluid flowrate on thermal resistance and temperature of the server for heat sink A is shown in Fig. 12 for both T and Z-type configurations. The reduction in thermal resistance with an increase in the fluid flowrate is qualitatively

similar to the reduction in the temperature of the server in all cases. Improvement in the performance of the system (i.e. reduction in both thermal resistance and temperature of server) with an increase in the fluid flowrate is significantly larger for PAO-6 compared to FC-40 owing to the larger viscosity of the former.

For T-type configuration with FC-40 as a working fluid, the thermal resistance of the system and temperature of the server decreases by around 10.9% and 4.3%, respectively when the flowrate is increased from 1 to 3 LPM, whereas for PAO-6, the reduction in the same is 21.1% and 11.9%, respectively. For Z-type configuration with FC-40 as a working fluid, the thermal resistance of the system and temperature of the server is decreased by around 0.4% and 0.3% respectively when the fluid flowrate is increased from 1 to 3

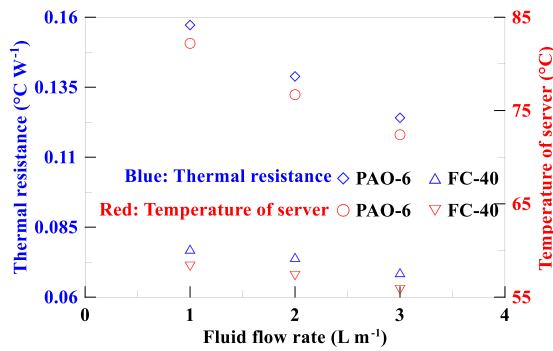


(a) Thermal resistance

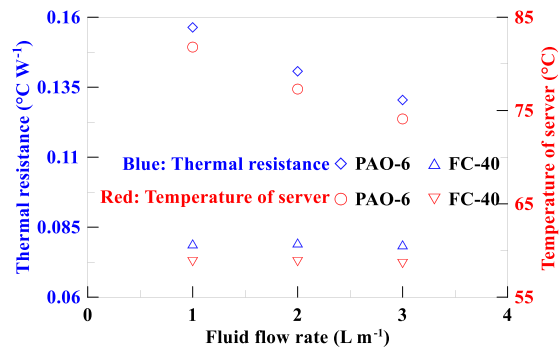


(a) Temperature of server

Fig. 11. Effect of suction on thermal resistance and temperature of server for T and Z-type configuration (No bypass of fluid, no suction fans, heat sink A with fin pitch of 1.2 mm, fluid flowrate: 3 LPM, fluid inlet temperature: 35 °C, and heat load: 300 W).

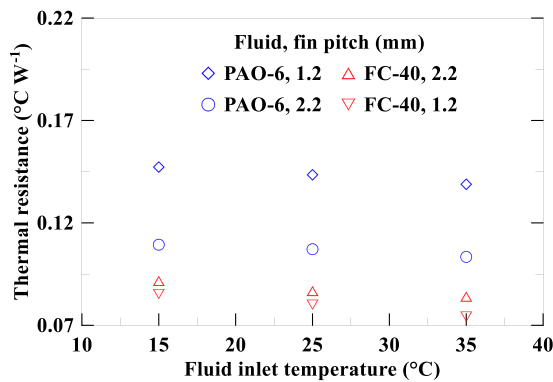


(a) T-type configuration

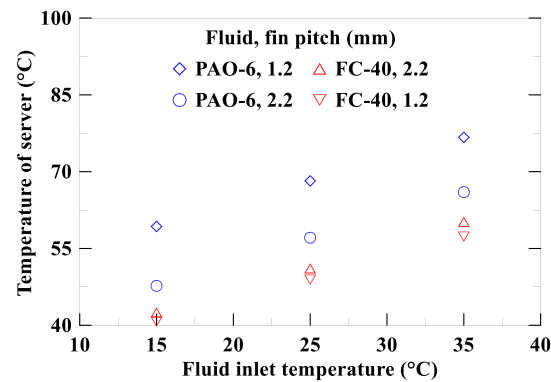


(b) Z-type configuration

Fig. 12. Effect of fluid flowrate on thermal resistance and temperature of server for T and Z-type configurations (No bypass of fluid, no suction fans, heat sink A with fin pitch of 1.2 mm, fluid inlet temperature: 35 °C, and heat load: 300 W).



(a) Thermal resistance



(b) Temperature of server

Fig. 13. Variation in thermal resistance and temperature of the server with fluid inlet temperature for heat sinks A and B (T-type configuration, no bypass of oil, no suction fans, flowrate: 2 LPM, fluid inlet temperature: 35 °C, and heat load: 300 W).

LPM, whereas for PAO-6, the reduction in the same is 16.6% and 9.4%, respectively. The reduction in thermal resistance and temperature shows monotonic tendency for all cases except for FC-40 with a Z-type configuration where the effect of fluid flowrate is almost negligible. This can be attributed to the comparatively smaller viscosity of FC-40 and the main flow did not enter the heat sink even though the flowrate is increased. Conversely, the much higher viscosity of PAO-6 causes more uniformity of the fluid flowrate in passing heat sinks and DDR ram module. In this regard,

the appreciable rise of the flowrate into the heat sink prevails with the increase of supplied flowrate, thus yielding a decrease in thermal resistance.

3.5. Fluid inlet temperature

The effect of fluid inlet temperature on the performance of the system employing both heat sinks A and B is shown in Fig. 13 depicting variation in thermal resistance and temperature of the

server. Performance of the system improves with an increase in the fluid inlet temperature for both fluids FC-40 and PAO-6 with both heat sinks A and B. Thermal resistance decreases and the temperature of the server increases monotonically with the fluid inlet temperature. Note that the effect of fin pitch on the variation of thermal resistance with inlet temperature is opposite for FC-40 and PAO-6. For PAO, thermal resistance is reduced when the fin pitch is increased from 1.2 mm to 2.2 mm. Conversely, the thermal resistance is increased. The opposite trend is primarily due to the difference in viscosity. The Effect of fluid inlet temperature on the temperature of the server is more significant compared to the thermal resistance. This can be attributed to the direct effect of the temperature of the coolant on heat transfer and hence the temperature of the server whereas thermal resistance is a derived quantity.

For heat sink A with FC-40 as a working fluid, the thermal resistance of the system decreases, and the temperature of the server increases by 12.9% and 40.8% respectively, whereas for PAO-6 the decrease and increase are 5.7% and 29.3% respectively. For heat sink B with FC-40 as a working fluid, the thermal resistance of the system decreases and the temperature of the server increases by 8.4% and 41.6% respectively, whereas for PAO-6 the decrease and increase are 5.4% and 38.4% respectively. The effect of fin pitch on the thermal performance is recorded opposite for both fluids similar to that reported while studying the effect of bypass in Section 3.2.

3.6. Heat load

Variations in the thermal resistance, temperature of the server, and temperature of the fluid at the outlet with the increase in the heat load for both heat sinks A and B are shown in Fig. 14. The thermal resistance monotonically decreases with the increase in the heat load for both the fluids. For heat sink A with FC-40 as a working fluid, the thermal resistance of the system decreases, and the temperature of the server increases by 10.4% and 49.5% respectively, whereas for PAO-6 the decrease and increase are 25.8% and 53.4% respectively. For heat sink B with FC-40 as a working fluid, the thermal resistance of the system decreases and the temperature of the server increases by 4.8% and 58.1% respectively, whereas for PAO-6 the decrease and increase are 11.3% and 59% respectively. A large decrease in the thermal resistance for heat sink A with PAO-6 as a working fluid can be attributed to a significant decrease in the viscosity of PAO-6 (around 19%) compared to that of FC-40 (12%) for an almost equal increase in the temperature of the fluid (around 6 °C). Note that the thermal resistance in a typical air-cooled heat sink is virtually unchanged with the supplied power, hence the change in the thermal resistance with heat load in the present immersion-cooled liquid-cooled heat sink is mainly associated with the property variation of viscosity. A decrease in the thermal resistance is more pronounced for PAO-6 compared to FC-40 especially for heat sink A with a fin pitch of 1.2 mm. For both fluids, heat sink A with a fin pitch of 1.2 mm leads to a larger reduction in the thermal resistance with an increase in the heat load. With the heat sink with a fin pitch of 1.2 mm, the viscosity of the fluid plays an important role in the overall heat transfer performance owing to smaller viscosity at larger heat loads and improved forced convection through the heat sink A. The decrease in thermal resistance can be attributed to reduced convection thermal resistance as a result of the improved flow of fluid around the server due to the significant decrease in viscosity as shown in Fig. 4(a). The temperature of the fluid at the outlet considerably increases with the increase in the heat load, especially for FC-40 due to lower specific heat compared to PAO-6.

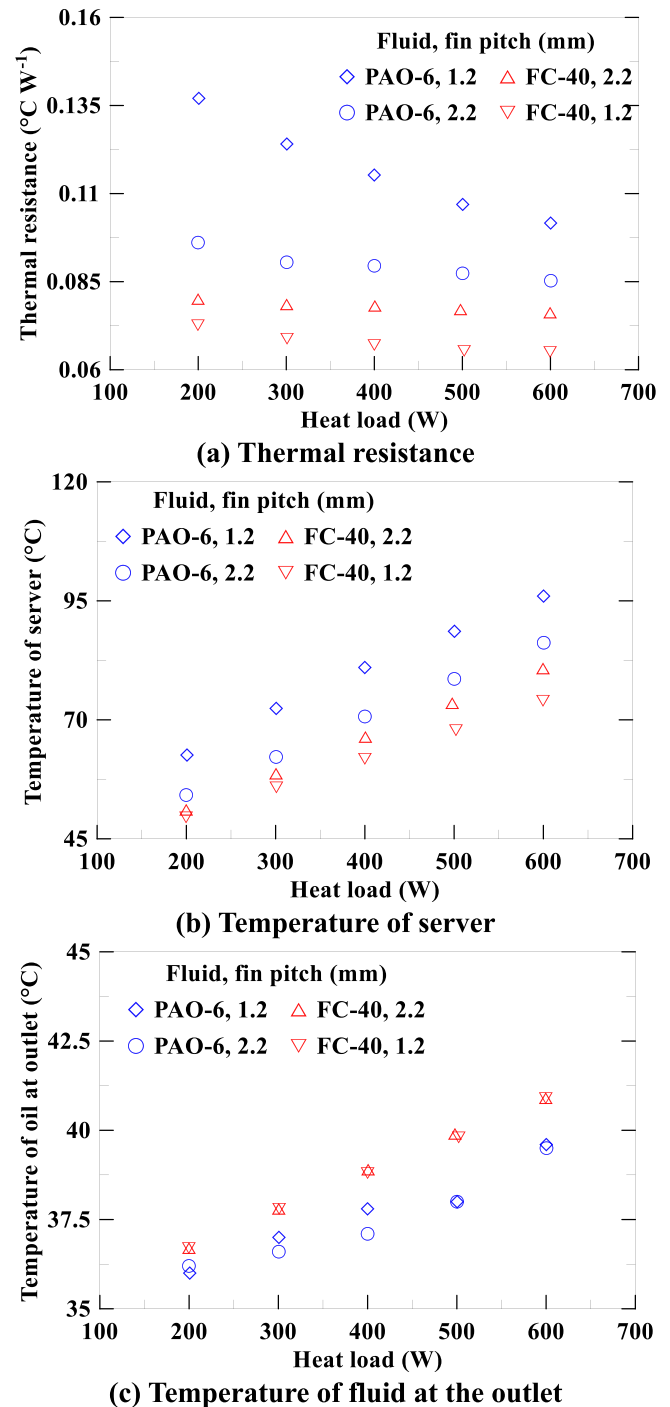


Fig. 14. Variations in thermal resistance, temperature of the server, and temperature of fluid at outlet with heat load (T-type configuration, no bypass of fluid, no suction fans, flowrate: 3 LPM, fluid inlet temperature: 35 °C, and heat load: 300 W).

The effect of fin pitch on the thermal performance is recorded opposite for both fluids similar to that reported while studying the effect of bypass (Section 3.2) and fluid inlet temperature (Section 3.6). Heat sink A offers roughly 19–43% more thermal resistance compared to heat sink B with PAO-6 as a working fluid whereas with FC-40, heat sink B offers around 10–17% more thermal resistance compared to heat sink A. Effect of the fin pitch is observed significantly for PAO-6 compared to FC-40 owing to larger resistance to flow (viscosity) compared to FC-40. Table 4 summa-

Table 4

Summary of the results from Sections 3.1–3.6.

Parameter (Range)		% variation in thermal resistance		% variation in temperature of server	
		FC-40	PAO-6	FC-40	PAO-6
Inlet/outlet configuration (T-type compared to Z or U-type)		−12.6	−5.1	−4.9	−2.3
Bypass of the fluid (gap: 0–5 mm)	Heat sink A	+6.9	+13.6	+2.6	+7.5
	Heat sink B	+2.9	+12.3	+1.2	+5.8
	T-type	−9.3	−15.6	−3.8	−8.3
	Z-type	−13	−16.7	−5.3	−8.8
Suction (no. of fans: 0–2)	T-type	−10.9	−21.1	−2.5	−11.9
	Z-type	−0.4	−16.6	−0.2	−9.4
Fluid flowrate (1–3 LPM)	Heat sink A	−12.9	−5.7	+40.8	+29.3
	Heat sink B	−8.4	−5.4	+41.6	+38.4
Fluid inlet temperature (15–35 °C)	Heat sink A	−10.4	−25.8	+49.5	+53.4
	Heat sink B	−4.8	−11.3	+58.1	+59

rized the parametric influences on the thermal resistance and temperature amid FC-40 and PAO-6 from Section 3.

4. Conclusions

The present work investigates experimentally and numerically the cooling of a 1 U server through single-phase immersion cooling. The working fluids include dielectric fluid FC-40 and dielectric oil PAO-6 with heat load spanning from 100 to 600 W and flowrate ranging from 1 to 3 LPM. The effects of configuration concerning the positions of the inlet and outlet, bypass of fluid, suction downstream the heat sink, design of heat sink, fluid flowrate, its inlet temperature, and heat load are studied on the performance of the system. Based on the foregoing discussion, the following conclusions are presented from the present work.

- 1 The lowest thermal resistance is achieved in the T-type configuration of the inlet and outlet. The effect of the configuration (or the orientation of the server) is insignificant for PAO-6 compared to FC-40.
- 2 Both the thermal resistance and temperature of the server increase monotonically with the bypass of fluid. The cooling performance of the system is improved when there is no bypass. The bypass of the fluid is reported to affect the performance of the system insignificantly compared to other parameters as a bypass of the fluid occurs pronouncedly in the small gap between the heat sink and tank.
- 3 Heat load affects the performance of the system significantly compared to other system parameters. Unlike typical air-cooled heat sinks, the thermal resistance subject to the single-phase immersion is reduced with the rise of heat load. However, the maximum temperature of the server also rises with the heat load. This is associated with the large decline in viscosity with temperature for the dielectric fluids. With the rise of heat load or temperature alongside the heat sink, the local variation in viscosity may become more pronounced, and it results in local flow mixing for considerable reduction of the viscosity. As a result, the resultant local turbulence provides better heat transfer performance.
- 4 Yet the PAO-6 shows an even more pronounced change in viscosity, thereby causing apparent fluid flow behavior across the heat sink. Large difference in fluid viscosity and variation with an increase in the temperature of fluid for both PAO-6 and FC-40 may highlight the difference in the heat transfer performance. The performance difference between the two heat sinks gradually decreases as the heat load increases.
- 5 Fluid inlet temperature is another dominant system parameter affecting the system performance having negative and positive correlations with thermal resistance and temperature of the server respectively similar to that by heat load. Large fluid

flowrate and placing suction improve the cooling performance. The effect of fluid flowrate on thermal resistance is even more pronounced for PAO-6 compared to FC-40.

- 6 With PAO-6 as a working fluid, a heat sink with a fin pitch of 2.2 mm performs better than that of 1.2 mm which is exactly opposite to FC-40. This can be attributed to a large difference in the viscosity of both fluids. Heat sink A with a fin pitch of 1.2 mm may offer PAO-6 a larger flow resistance across the heat sink, thereby raising higher thermal resistance larger resistance to the conduction heat transfer (with thermal conductivity roughly 2700 times smaller compared to copper) compared to heat sink B with fin pitch of 2.2 mm.

Declaration of Competing Interest

I am the corresponding author (Chi-Chuan Wang) and declare no conflict of interest of this submission.

CRediT authorship contribution statement

Harshad Shrigondekar: Conceptualization, Writing – original draft. **Yueh-Cheng Lin:** Investigation, Writing – original draft. **Chi-Chuan Wang:** Conceptualization, Supervision, Writing – review & editing.

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References

- [1] A. Andrae, T. Edler, On global electricity usage of communication technology: trends to 2030, *Challenges* 6 (2015) 117–157.
- [2] E. Masanet, A. Shehabi, N. Lei, S. Smith, J. Koomey, Recalibrating global data center energy-use estimates, *Science* 367 (2020) 984–986.
- [3] Z. Han, X. Sun, H. Wei, Q. Ji, D. Xue, Energy saving analysis of evaporative cooling composite air conditioning system for data centers, *Appl. Therm. Eng.* 186 (2021) 116506.
- [4] Z. He, T. Ding, Y. Liu, Z. Li, Analysis of a district heating system using waste heat in a distributed cooling data center, *Appl. Therm. Eng.* 141 (2018) 1131–1140.
- [5] Y. Fulpagare, A. Bhargav, Advances in data center thermal management, *Renew. Sustain. Energy Rev.* 43 (2015) 981–996.
- [6] S. Alkharabsheh, J. Fernandes, B. Gebrehiwot, D. Agonafer, K. Ghose, A. Ortega, Y. Joshi, B. Sammakia, A brief overview of recent developments in thermal management in data centers, *J. Electron. Packag.* (2015) 137.
- [7] A.H. Khalaj, S.K. Halgamuge, A Review on efficient thermal management of air-and liquid-cooled data centers: from chip to the cooling system, *Appl. Energy* 205 (2017) 1165–1188.
- [8] A.C. Kheirabadi, D. Groulx, Cooling of server electronics: a design review of existing technology, *Appl. Therm. Eng.* 105 (2016) 622–638.

- [9] H.M. Daraghme, C.C. Wang, A review of current status of free cooling in data-centers, *Appl. Therm. Eng.* 114 (2017) 1224–1239.
- [10] W.X. Chu, C.C. Wang, A review on airflow management in data centers, *Appl. Energy* 240 (2019) 84–119.
- [11] Y. Joshi, P. Kumar, *Energy Efficient Thermal Management of Data Centers*, Springer Science & Business Media, 2012.
- [12] S.P. Gurrum, S.K. Suman, Y.K. Joshi, A.G. Fedorov, Thermal issues in next-generation integrated circuits, *IEEE Trans. Device Mater. Reliab.* 4 (2004) 709–714.
- [13] K.J. Kim, H.H. Yang, W. Noh, B. Han, A. Bar-Cohen, Thermal performance of cryogenic micro-pin fin coolers with two-phase liquid nitrogen flows, *Appl. Sci.* 11 (2021) 11071.
- [14] A. Reeser, A. Bar-Cohen, G. Hetsroni, High quality flow boiling heat transfer and pressure drop in microgap pin fin arrays, *Int. J. Heat Mass Transf.* 78 (2014) 974–985.
- [15] A. Gorasiya, S.K. Saha, Single phase laminar fluid flow and heat transfer in microchannel with cylindrical and parallelepiped micro-fins, *Heat Mass Transf.* 55 (3) (2019) 613–626.
- [16] A. Bar-Cohen, J.J. Maurer, D.H. Altman, Embedded cooling for wide bandgap power amplifiers: a review, *J. Electron. Packag.* 141 (4) (2019) 040803.
- [17] M. Misale, J.A. Bocanegra, A. Marchitto, Thermo-hydraulic performance of connected single-phase natural circulation loops characterized by two different inner diameters, *Int. Commun. Heat Mass Transf.* 125 (2021) 105309.
- [18] J.A. Bocanegra, A. Marchitto, M. Misale, Thermal performance investigation of a mini natural circulation loop for solar PV panel or electronic cooling simulated by lattice Boltzmann method, *Int. J. Energy Prod. Manag.* 7 (1) (2022) 1–12.
- [19] M. Saini, R.L. Webb, Heat rejection limits of air cooled plane fin heat sinks for computer cooling, in: *Proceedings of the iTherm 8th Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems* (Cat. No. 02CH37258), IEEE, 2002, pp. 1–8.
- [20] S. Fan, F. Duan, A review of two-phase submerged boiling in thermal management of electronic cooling, *Int. J. Heat Mass Transf.* (2020) 150.
- [21] I. Levin, A. Dordopulo, Y. Doronchenko, M. Raskladkin, A. Fedorov, Z. Kalyaev, Immersion liquid cooling FPGA-based reconfigurable computer system, *IFAC-PapersOnLine* 49 (2016) 366–371.
- [22] I. Kuncoro, N. Pambudi, M. Biddinika, I. Widiastuti, M. Hijriawan, K. Wibowo, Immersion cooling as the next technology for data center cooling: a review, *J. Phys. Conf. Ser.* 1402 (2019) 044057 IOP Publishing.
- [23] E. Baker, Liquid immersion cooling of small electronic devices, *Microelectron. Reliab.* 12 (1973) 163–173.
- [24] B.B. Kanbur, C. Wu, S. Fan, F. Duan, System-level experimental investigations of the direct immersion cooling data center units with thermodynamic and thermoeconomic assessments, *Energy* 217 (2021).
- [25] A. ASHRAE, Standard 34-2019 Designation and Safety Classification of Refrigerants, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc., Atlanta, GA, USA, 2019.
- [26] S. Ramdas, P. Rajmane, T. Chauhan, A. Misrak, D. Agonafer, Impact of immersion cooling on thermo-mechanical properties of PCB's and reliability of electronic packages, in: *Proceedings of the International Electronic Packaging Technical Conference and Exhibition*, 59322, American Society of Mechanical Engineers, 2019 pp. V001T002A011.
- [27] J.M. Shah, R. Eiland, A. Siddharth, D. Agonafer, Effects of mineral oil immersion cooling on IT equipment reliability and reliability enhancements to data center operations, in: *Proceedings of the 15th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm)*, IEEE, 2016, pp. 316–325.
- [28] Q. Luo, C. Wang, H. Wen, L. Liu, Research and optimization of thermophysical properties of sic oil-based nanofluids for data center immersion cooling, *Int. Commun. Heat Mass Transf.* (2022) 131.
- [29] R. Eiland, J. Fernandes, M. Vallejo, D. Agonafer, V. Mulay, Flow Rate/Flowrate and inlet temperature considerations for direct immersion of a single server in mineral oil, in: *Proceedings of the 14th Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm)*, IEEE, 2014, pp. 706–714.
- [30] P.V. Bansode, J.M. Shah, G. Gupta, D. Agonafer, H. Patel, D. Roe, R. Tufty, Measurement of the thermal performance of a single-phase immersion cooled server at elevated temperatures for prolonged time, in: *Proceedings of the International Electronic Packaging Technical Conference and Exhibition*, 51920, American Society of Mechanical Engineers, 2018 pp. V001T002A010.
- [31] P.V. Bansode, J.M. Shah, G. Gupta, D. Agonafer, H. Patel, D. Roe, R. Tufty, Measurement of the thermal performance of a custom-build single-phase immersion cooled server at various high and low temperatures for prolonged time, *J. Electron. Packag.* (2020) 142.
- [32] D. Gandhi, U. Chowdhury, T. Chauhan, P. Bansode, S. Saini, J.M. Shah, D. Agonafer, Computational analysis for thermal optimization of server for single phase immersion cooling, in: *Proceedings of the ASME International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems*, 2019 Vol. ASME 2019 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems.
- [33] P.A. Shinde, P.V. Bansode, S. Saini, R. Kasurthy, T. Chauhan, J.M. Shah, D. Agonafer, Experimental analysis for optimization of thermal performance of a server in single phase immersion cooling, in: *Proceedings of the International Electronic Packaging Technical Conference and Exhibition*, 59322, American Society of Mechanical Engineers, 2019 pp. V001T002A014.
- [34] C.C. Cheng, P.C. Chang, H.C. Li, F.I. Hsu, Design of a single-phase immersion cooling system through experimental and numerical analysis, *Int. J. Heat Mass Transf.* (2020) 160.
- [35] M. Matsuoka, K. Matsuda, H. Kubo, Liquid immersion cooling technology with natural convection in data center, in: *Proceedings of the IEEE 6th International Conference on Cloud Networking (CloudNet)*, IEEE, 2017, pp. 1–7.
- [36] J.M. Shah, R. Dandamudi, C. Bhatt, P. Rachamreddy, P. Bansode, D. Agonafer, CFD analysis of thermal shadowing and optimization of heatsinks in 3rd generation open compute server for single-phase immersion cooling, in: *Proceedings of the International Electronic Packaging Technical Conference and Exhibition*, 59322, American Society of Mechanical Engineers, 2019 pp. V001T002A015.
- [37] J.M. Shah, C. Bhatt, P. Rachamreddy, R. Dandamudi, S. Saini, D. Agonafer, Computational form factor study of a 3rd generation open compute server for single-phase immersion cooling, in: *Proceedings of the International Electronic Packaging Technical Conference and Exhibition*, 59322, American Society of Mechanical Engineers, 2019 pp. V001T002A017.
- [38] M. Misale, F. Devia, P. Garibaldi, Experiments with Al2O3 nanofluid in a single-phase natural circulation mini-loop: preliminary results, *Appl. Therm. Eng.* 40 (2012) 64–70.
- [39] A. Niazmand, P. Murthy, S. Saini, P. Shahi, P. Bansode, D. Agonafer, Numerical analysis of oil immersion cooling of a server using mineral oil and Al2O3 nanofluid, in: *Proceedings of the International Electronic Packaging Technical Conference and Exhibition*, 84041, American Society of Mechanical Engineers, 2020 pp. V001T008A009.
- [40] M. Muneeshwaran, YC Lin, CC Wang, Performance analysis of single-phase immersion cooling system of data center using FC-40 dielectric fluid, *International Communications in Heat and Mass Transfer* (Under Review)
- [41] H.W. Coleman, W.G. Steele, *Experimentation, Validation, and Uncertainty Analysis For Engineers*, John Wiley & Sons, 2018.